



NEET TEST HUB

NTH-PREMIUM TEST SERIES 13

Date – 04 January 2026

Duration – 3 Hours

Marks – 720

Physics :- Ray Optics & Optical Instruments, Wave Optics

Chemistry :- Organic Chemistry: Some Basic Principles & Techniques, Haloalkanes & Haloarenes, Alcohol, Phenol & Ether

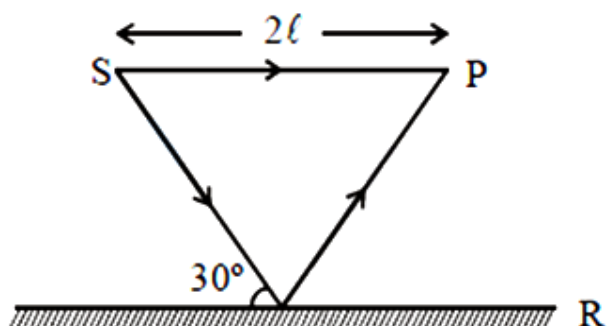
Biology :- Human Health And Disease, Microbes in Human Welfare, Biotechnology: Principles & Processes, Biotechnology & its Applications

Instruction

1. The test is of 3 hours duration.
2. The test booklet consists of 200 questions. The maximum mark is 720.
3. There are four Sections in the Question Paper, Sections I, II, III, and IV consisting of Section I (Physics), Section II (Chemistry), Section III (Botany) and Section IV (Zoology) have 50 Questions in each Subject and each subject is divided into two Sections,
4. Section A consists of 35 questions (all questions compulsory) and Section B consists of 15 Questions (Any 10 questions are compulsory).
5. There is only one correct response for each question.
6. Each correct answer will give 4 marks while 1 Mark will be deducted for a wrong MCQ response.
7. No student is allowed to carry any textual material, printed, or written, bits of paper, pager, mobile phone, any electronic device, etc. Inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the Candidates are allowed to take away this Test Booklet with them

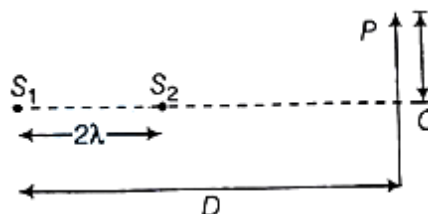
Physics

1. A double-slit experiment is immersed in a liquid of refractive index 1.33. The separation between the slits is 1 mm and the distance between the slit and screen is 1.33 m. If slits are illuminated by a parallel beam of light whose wavelength is $6300 \text{ \AA}$, then fringe width will be
 A) 6.3 mm B) 63 mm
 C) 0.63 mm D) None of these
2. If the fringe width obtained in Young's double-slit experiment is 1.33 mm, then on immersing the whole set up in the water with refractive index 1.33, new fringe width will be
 A) Does not change B) 1.0 mm
 C) 2.0 mm D) 2.69 mm
3. In Young's double-slit experiment, the fringe width is found to be 0.4 mm. If the whole apparatus is immersed in water of refractive index $\frac{4}{3}$, without disturbing the geometrical arrangement, the new fringe width will be
 A) 0.30 mm B) 0.40 mm
 C) 0.53 mm D) 450 microns
4. The interference pattern is observed at P due to superimposition of two rays coming out from a source S as shown in the figure. The value of l for which maxima is obtained at P is (R is a perfect reflecting surface): -



- A) $l = \frac{(2n-1)\lambda}{2(\sqrt{3}-1)}$ B) $l = \frac{(2n-1)\lambda}{\sqrt{3}-1}$
 C) $l = \frac{(2n-1)\lambda\sqrt{3}}{4(2-\sqrt{3})}$ D) $l = \frac{2n\lambda}{\sqrt{3}-1}$

5. Two coherent point sources S_1 and S_2 vibrating in phase emit light of wavelength λ . The separation between them is 2λ . The light is collected on a screen placed at a distance $D > > \lambda$ from the slit S_1 as shown. The minimum distance, so that intensity at P is equal to the intensity at O



- A) $\sqrt{2}D$ B) $\sqrt{3}D$ C) $\sqrt{8}D$ D) $\sqrt{5}D$
6. In Young's experiment, the wavelength of red light is $7.8 \times 10^{-8} \text{ cm}$ and that of blue light is $5.2 \times 10^{-8} \text{ cm}$. Value of n for which $(n+1)^{\text{th}}$ blue bright fringe coincides with n^{th} red fringe is
 A) 1 B) 2 C) 3 D) 4
 7. In Young's double-slit experiment, the fringe width is found to be 0.4 mm. If the whole apparatus is immersed in water of refractive index $\frac{4}{3}$, without disturbing the geometrical arrangement, the new fringe width will be
 A) 0.30 mm B) 0.40 mm
 C) 0.53 mm D) 450 microns
 8. A parallel beam of monochromatic light falls normally on a single narrow slit. How does the angular width of the principal maximum in the resulting diffraction pattern varies with the width of the slit?
 A) Principal maximum increases on increasing the width of slit
 B) Principal maximum decreases on decreasing the width of slit
 C) Principal maximum increases on decreasing the width of slit
 D) None of these
 9. Unpolarized light passes through two polaroids, the axis of one is vertical and that of the other is 30° to the vertical. What is the orientation and intensity of the transmitted light?
 A) Plane polarized at 60° to the vertical and having intensity of $\frac{I_0}{4}$

- B) Plane polarized at 30° to the vertical and having intensity of $\frac{3I_0}{8}$
 C) Plane polarized at 30° to the vertical and having intensity of $\frac{I_0}{2}$
 D) No light passes
10. Two coherent point sources S_1 and S_2 vibrating in phase emit light of wavelength λ . The separation between the sources is 2λ . The smallest distance from S_2 on a line passing through S_2 and perpendicular to S_1S_2 where a minimum intensity occurs is
 A) $\frac{7\lambda}{12}$ B) $\frac{15\lambda}{4}$ C) $\frac{\lambda}{2}$ D) $\frac{3\lambda}{4}$
11. Unpolarized light of intensity 32 W m^{-2} passes through three polarizers such that transmission axes of the first and second polarizer make an angle 30° with each other and the transmission axis of the last polarizer is crossed with that of the first. The intensity of the final emerging light will be
 A) 32 W m^{-2} B) 3 W m^{-2}
 C) 8 W m^{-2} D) 4 W m^{-2}
12. If yellow light emitted by sodium lamp in Young's double-slit experiment is replaced by a monochromatic blue light of the same intensity
 A) fringe width will decrease
 B) fringe width will increase
 C) fringe width will remain unchanged
 D) fringe width will become less intense
13. Unpolarized light of intensity I passes through an ideal polarizer A. Another identical polarizer B is placed behind A. The intensity of light beyond B is found to be $\frac{I}{2}$. Now another identical polarizer C is placed between A and B. The intensity beyond B is now found to be $\frac{I}{8}$. The angle between polarizer A and C is
 A) 60° B) 0° C) 30° D) 45°
14. In Young's double-slit experiment, in an interference pattern, the second minimum is observed exactly in front of one slit. The distance between the slits is d and the distance between source and screen is D . The wavelength of the light source used is
 A) $\frac{d^2}{D}$ B) $\frac{d^2}{2D}$ C) $\frac{d^2}{3D}$ D) $\frac{d^2}{4D}$
15. In Young's double-slit experiment, the separation between the slits is halved and the distance between the slits and the screen is doubled. The fringe width will
 A) remain the same B) be halved
 C) be doubled D) be quadrupled

16. In Young's double-slit experiment, the number of maxima that are obtained on a screen (including the central maxima), if $\lambda = 2000 \text{ \AA}$ and $d = 7000 \text{ \AA}$ is

A) 12 B) 7 C) 18 D) 6

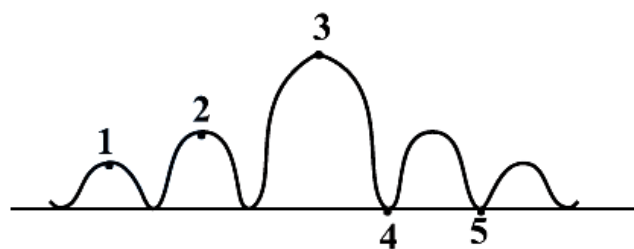
17. What happens to the fringe pattern if in the path of one of the slits a glass plate which absorbs 50% energy is introduced?

A) The bright fringes become brighter and dark fringes become darker
 B) No fringes are observed
 C) The fringe width decreases
 D) None of the above

18. A telescope of objective lens diameter 2 m uses light of wavelength $5000 \text{ \AA}$ for viewing stars. The minimum angular separation between two stars whose image is just resolved by the telescope is

A) $4 \times 10^{-4} \text{ rad}$ B) $0.25 \times 10^{-6} \text{ rad}$
 C) $0.31 \times 10^{-6} \text{ rad}$ D) $5 \times 10^{-3} \text{ rad}$

19. The diffraction pattern of a single slit is shown in the figure. The point at which the path difference of the extreme rays is two times the wavelength is



A) point 1 B) point 2 C) point 4 D) point 5

20. In single slit diffraction of light of wavelength λ by a slit of width e , the size of the central maximum on a screen at a distance b is

A) $2b\lambda + e$ B) $\frac{2b\lambda}{e}$
 C) $\frac{2b\lambda}{e} + e$ D) $\frac{2b\lambda}{e} - e$

21. In a system of two polarisers, it is found that the intensity of light from the second polarizer is half from that of the first polariser. The angle between their pass axes is

A) 60° B) 30° C) 0° D) 45°

22. In Young's double slits experiment, if the separation between the slits is halved, and the distance between the slits and the screen is doubled, then the fringe width compared to the original one will be

A) Unchanged B) Halved C) Doubled D) Quadrupled

23. A plane mirror of circular shape with radius $r = 20$ cm is fixed to the ceiling. A bulb is to be placed on the axis of the mirror. A circular area of the radius $R = 1$ m on the floor is to be illuminated after reflection of light from the mirror. The height of the room is 3 m. What is the maximum distance from the centre of the mirror and the bulb so that the required area is illuminated?

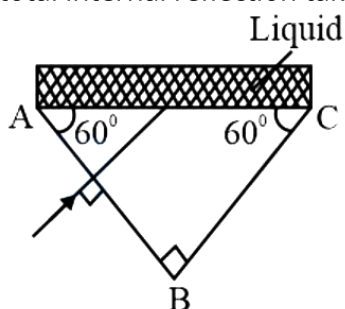
A) 75 cm B) 70 cm
C) 80 cm D) 100 cm

24. The frequency of a light wave in a material is 2×10^{14} Hz and wavelength is $5000 \text{ \AA}$. The refractive index of material will be

A) 1.40 B) 1.50 C) 3.00 D) 1.33

25.

Light is incident normally on the face AB of a prism as shown in the figure. A layer of liquid of refractive index μ is applied on the face AC of the prism. If the prism is made of glass of refractive index $\frac{3}{2}$, then the limit of μ for which total internal reflection takes place on face AC is



A) $\mu < \frac{3\sqrt{3}}{4}$ B) $\mu > \sqrt{\frac{5}{2}}$ C) $\mu > \sqrt{2}$ D) $\mu < \sqrt{3}$

26. Laser light has the following property

A) Laser light always lies in the X-ray region
B) Laser light does not have directionality property
C) Laser light is white light
D) Laser light is highly coherent

27. Two lenses of power +10 D and -5 D are placed in contact. Where should an object be held from the lens, so as to obtain a virtual image of magnification 2?

A) 5 cm B) -5 cm
C) 10 cm D) -10 cm

28. A convex, rearview mirror of focal length 20 cm, is fitted in a car. A second car 2 m broad and 1.6 m high is 6 m away from the first car and overtakes the first car at a relative speed of 15 m s^{-1} , then the speed of the image of the second car with respect to the first car is

A) 0.016 m s^{-1}
B) 0.0257 m s^{-1}
C) 0.162 m s^{-1}
D) 0.0073 m s^{-1}

29. A ray of light strikes a transparent slab of the refractive index of $\sqrt{2}$ at an angle of incidence 45° . The angle between the reflected and refracted rays is

A) 95° B) 120°
C) 135° D) 105°

30. The red light of a wavelength $6400 \text{ \AA}$ in the air has a wavelength of $4000 \text{ \AA}$ in glass. If the wavelength of violet light in air is $4400 \text{ \AA}$, then the wavelength in the glass is

A) $2570 \text{ \AA}$ B) $2750 \text{ \AA}$
C) $1600 \text{ \AA}$ D) $2510 \text{ \AA}$

31. Magnification of a compound microscope is 30. The focal length of eye-piece is 5 cm and the image is formed at a distance of 'distinct vision' of 25 cm. Magnification of the objective lens is

A) 6 B) 5 C) 7.5 D) 10

32. For a prism of refractive index, 1.732, the angle of minimum deviation is equal to the angle of prism. Then the angle of the prism is

A) 60° B) 70°
C) 50° D) none of these

33. Near and far points of human eye are

A) 25 cm and infinite
B) 25 cm and 100 cm
C) 55 cm and 200 cm
D) 0 cm and 25 cm

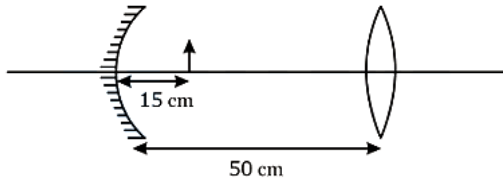
34. In a compound microscope, the focal lengths of two lenses are 1.5 cm and 6.25 cm. An object is placed at 2 cm from objective and the final image is formed at 25 cm from eye lens. The distance between the two lenses is

A) 6.00 cm B) 7.75 cm C) 9.25 cm D) 11.00 cm

35. In a compound microscope, the focal lengths of two lenses are 1.5 cm and 6.25 cm and an object is placed at 2 cm from objective and the final image is formed at 25 cm from eye lens. The distance between the two lenses is

A) 6.00 cm B) 7.75 cm C) 9.25 cm D) 11.00 cm

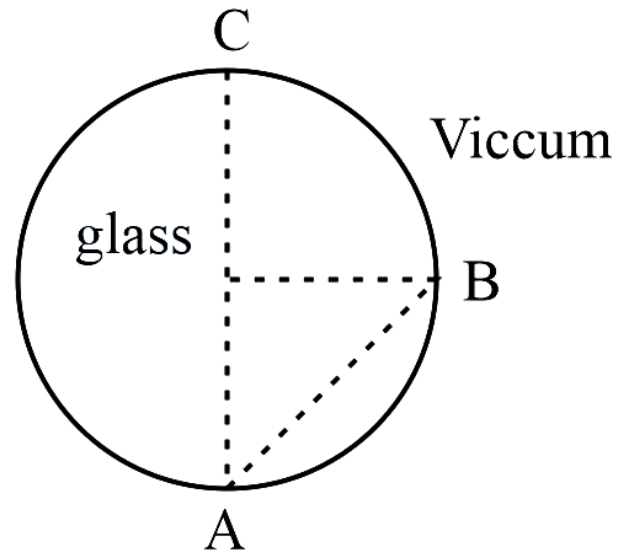
36. Consider a concave mirror and a convex lens (refractive index = 1.5) of focal length 10 cm each, separated by a distance of 50 cm in air (refractive index = 1), as shown in the figure. An object is placed at a distance of 15 cm from the mirror. Its erect image, formed by this combination, has magnification M_1 . When the set-up is kept in a medium of refractive index $\frac{7}{6}$, the magnification becomes M_2 . The magnitude $\left| \frac{M_2}{M_1} \right|$ is



A) 7 B) 5 C) 3 D) 6

37. An equilateral prism deviates a ray through 45° for two angles of incidence differing by 20° . The angles of incidence are :
- A) 60° and 40°
 B) 50° and 70°
 C) $62^\circ 30'$ and $42^\circ 30'$
 D) $60^\circ 30'$ and $40^\circ 30'$
38. If the angle of minimum deviation is of 60° for an equilateral prism, then the refractive index of the material of the prism is
- A) 1.41 B) 1.5 C) 1.6 D) 1.73
39. Time taken by the sunlight to pass through a slab of 4 cm and refractive index 1.5 is ____ s.
- A) 2×10^{10} B) 2×10^{-8} C) 2×10^8 D) 2×10^{-10}

40. It is found that all electromagnetic signals sent from A towards B reach point C. The speed of electromagnetic signals in glass cannot be

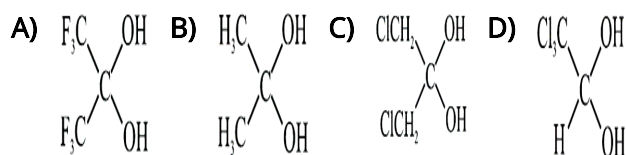


A) $1.0 \times 10^8 \text{ m s}^{-1}$
 B) $2.15 \times 10^8 \text{ m s}^{-1}$
 C) $2 \times 10^7 \text{ m s}^{-1}$
 D) $4 \times 10^7 \text{ m s}^{-1}$

41. The refractive index of water is 1.33. What will be the speed of light in water?
- A) $3 \times 10^8 \text{ ms}^{-1}$
 B) $4 \times 10^8 \text{ ms}^{-1}$
 C) $1.33 \times 10^8 \text{ ms}^{-1}$
 D) $2.26 \times 10^8 \text{ ms}^{-1}$
42. A ray of light passing through a prism of refractive index $\sqrt{2}$ undergoes minimum deviation. It is found that the angle of incidence is double the angle of refraction within the prism. The angle of the prism is
- A) 60° B) 90° C) 75° D) 30°
43. A wave has velocity u in medium P and velocity $2u$ in medium Q . If the wave is refracted from medium P to medium Q at an angle of incidence 30° , then the angle of refraction will be
- A) 30° B) 45° C) 60° D) 90°
44. A double convex thin lens made of the refractive index 1.6 has radii of curvature 15 cm each. The focal length of this lens when immersed in a fluid of refractive index 1.63, is
- A) 25 cm B) 125 cm
 C) 250 cm D) -407.5 cm
45. A thin rod of 5 cm length is kept along the axis of a concave mirror of 10 cm focal length such that its image is real and magnified and one end touches the rod. Its magnification will be

Chemistry

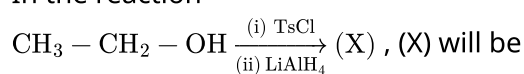
46. Which of the following geminal diols is most unstable:



47. The compound formed when Ethyl bromide is heated with dry Silver oxide is

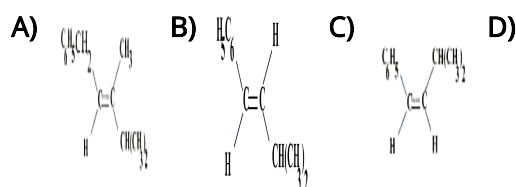
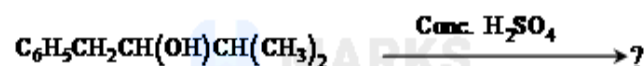
- A) dimethylether B) diethylether
C) methylalcohol D) ethylalcohol

48. In the reaction



- A) $\text{CH}_2 = \text{CH}_2$ B) $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{CH}_3$
C) $\text{CH}_3 - \text{CH}_3$ D) $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$

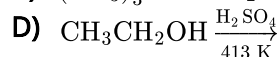
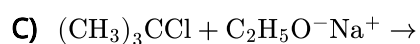
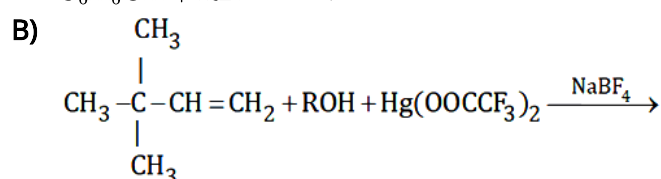
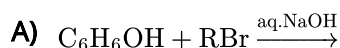
49. The main product of the following reactions is



50. Which of the compounds HCHO (I), $\text{CH}_3\text{CH}_2\text{CHO}$ (II), CH_3COCH_3 (III) and HCOOC_2H_5 (IV) will give a secondary alcohol on reaction with excess Grignard reagent, followed by hydrolysis? Select the correct answer using the codes given below

- A) II only B) III only C) I and IV D) II and IV

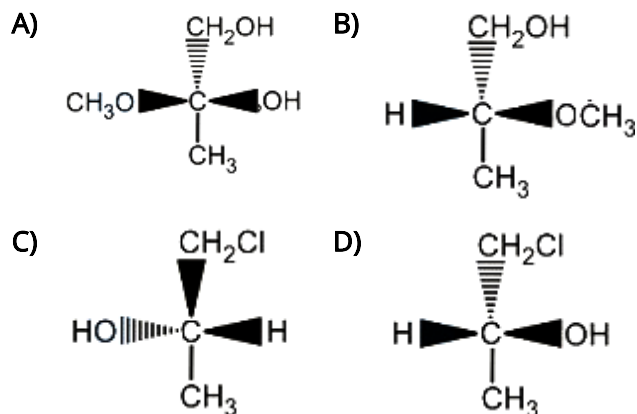
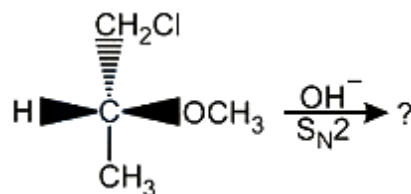
51. Which of the following reaction does not form an ether?



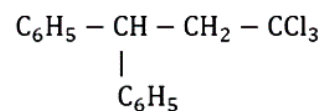
52. Which of the following can work as a dehydrating agent for alcohols

- A) H_2SO_4 B) Al_2O_3 C) H_3PO_4 D) All of these

53. What is the major product of the reaction ?



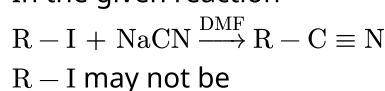
54.



The IUPAC name is

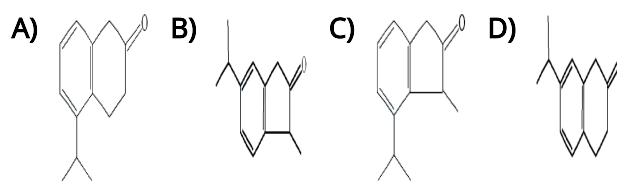
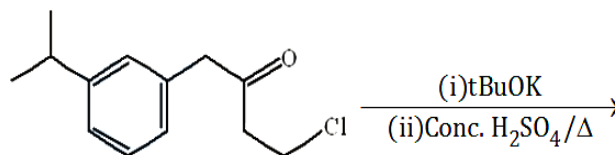
- A) 1, 1, 1-trichloro-3, 3-diphenyl propane
B) 1, 1-diphenyl-3, 3, 3-trichloro propane
C) Both (A) and (B)
D) None of these

55. In the given reaction

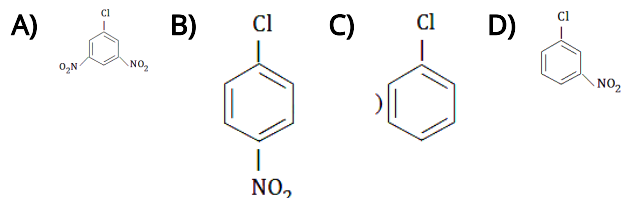


- A) Primary B) Secondary C) Tertiary D) $\text{CH}_3 - \text{I}$

56. The major product of the following reaction is



57. Which of the aromatic compounds react fastest with methoxide ion?

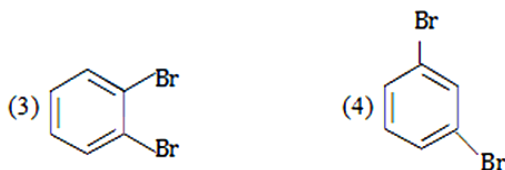


58. The addition of Br_2 to (E)-but-2-ene gives

- A) (R, R) – 2 – 3 – dibromobutane
 B) (S, R) – 2 – 3 – dibromobutane
 C) (R, S) – 2 – 3 – dibromobutane
 D) A mixture of (R, R) and (S, S) 2, 3 dibromobutane

59. Which of the following will not form Grignard reagent with Mg/dry ether?

- (1) $\text{CH}_2\text{Br}-\text{CH}_2\text{Br}$ (2) $\text{CH}_2\text{Br}-\text{CH}_2-\text{CH}_2\text{Br}$



Select the correct answer using the codes given below

- A) Only (4) B) Only (2)
 C) 1, 2 and 3 D) 1, 2, 3 and 4

60. How is the relative adsorption of each component of the mixture expressed in thin layer chromatography?

- A) By their color
 B) By their boiling points
 C) By their density
 D) By their retardation factor (R_f) value

61. What is the common suffix for the names of straight chain saturated hydrocarbons?

- A) -ene B) -ane C) -ol D) -ine

62. What is the relationship between hybridization and bond length?

- A) Hybridization has no effect on bond length.
 B) sp hybridization results in shorter bonds than sp^2 and sp^3 hybridization.
 C) sp^3 hybridization results in shorter bonds than sp and sp^2 hybridization.
 D) sp^2 hybridization results in shorter bonds than sp and sp^3 hybridization.

63. How does the relative electronegativity of carbon atoms with different hybrid orbitals affect the properties of molecules?

- A) It has no effect on the properties of molecules.
 B) It affects the color of the molecules.

- C) It influences the physical and chemical properties of the molecules.
 D) It changes the odor of the molecules.

64. What is necessary for proper sideways overlap in π bond formation?

- A) Perpendicular orientation of p orbitals
 B) Parallel orientation of p orbitals
 C) s orbital overlap
 D) d orbital overlap

65. Which atoms are specifically written in bond-line structural representations?

- A) Carbon and hydrogen atoms
 B) Oxygen, chlorine, and nitrogen atoms
 C) Hydrogen atoms only
 D) All atoms except carbon and hydrogen

66. Who synthesized acetic acid in 1845, demonstrating that organic compounds could be synthesized from inorganic sources in a laboratory?

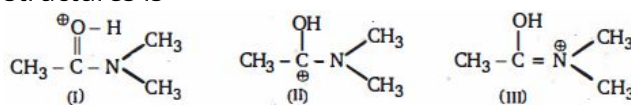
- A) F. Wohler B) Berthel C) Berzilius D) Kolbe

67. Decreasing order of stability of following carbocations is

- (A) $m - \text{CH}_3\text{OPhCH}_2^+$ (B) $p - \text{CH}_3\text{OPhCH}_2^+$
 (C) PhCH_2^+ (D) $p - \text{NO}_2\text{PhCH}_2^+$

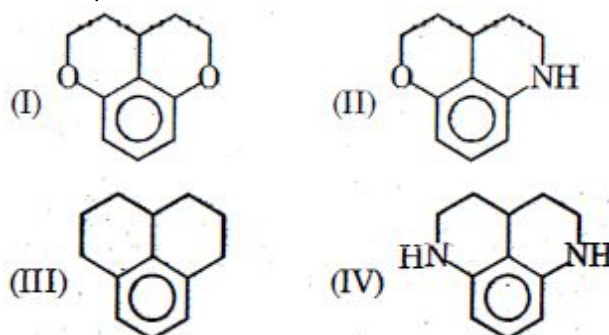
- A) $A > B > C > D$
 B) $B > C > D > A$
 C) $C > B > A > D$
 D) $B > C > A > D$

68. The correct stability order of the given canonical structures is



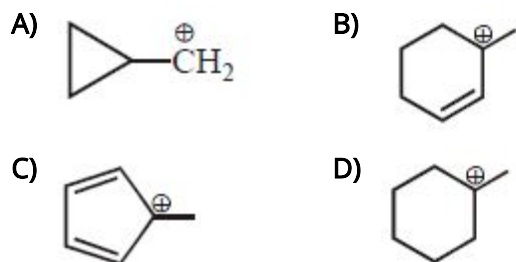
- A) $I > II > III$
 B) $III > I > II$
 C) $I > III > II$
 D) $II > III > I$

69. In above compounds, decreasing order of rate of electrophilic aromatic substitution will be ?

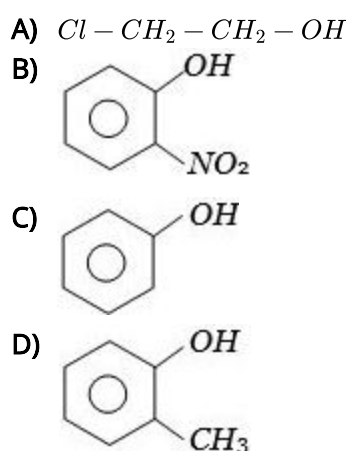


- A) $I > II > III > IV$
 B) $III > IV > II > I$
 C) $IV > II > I > III$
 D) $IV > I > II > III$

70. Identify the correct order of stability of carbocation



71. Which one of the following compounds is most acidic



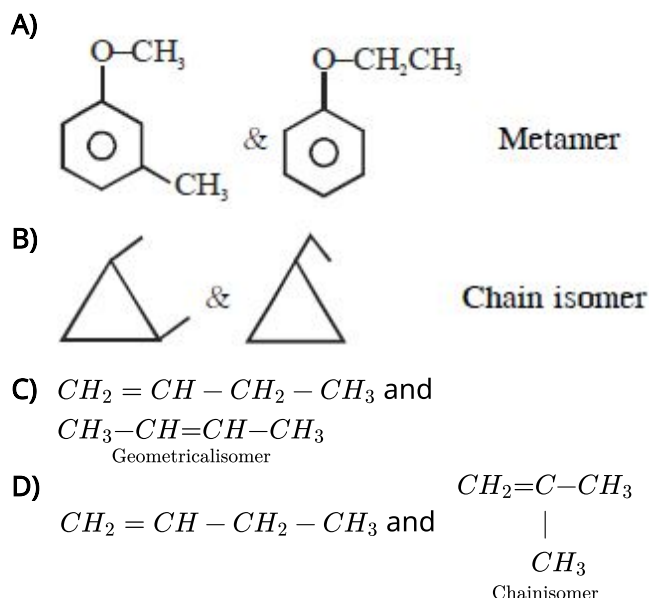
72. Which behaves both as a nucleophile and electrophile



73. In the mixture of conc. H_2SO_4 and HNO_3 the nitrating species is



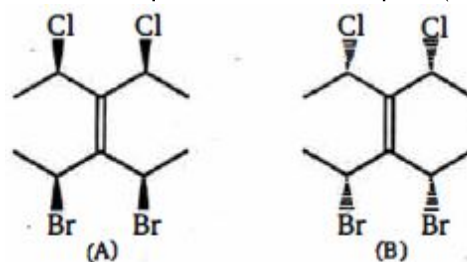
74. Which of the following is not correctly matched



75. Maleic acid and fumaric acids are

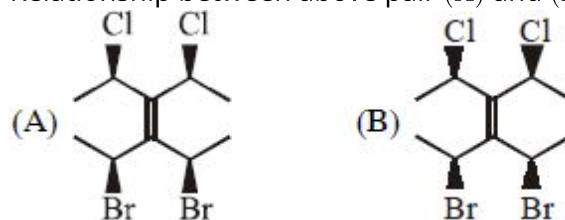
- A) Chain isomers
 B) Functional isomers
 C) Tautomers
 D) Geometrical isomers

76. Relationship between above pair (A) & (B) is



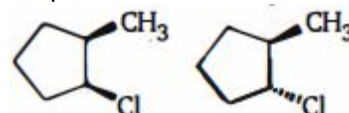
- A) Enantiomer B) Diastereomers
 C) Identical D) Structural isomer

77. Relationship between above pair (A) and (B) is



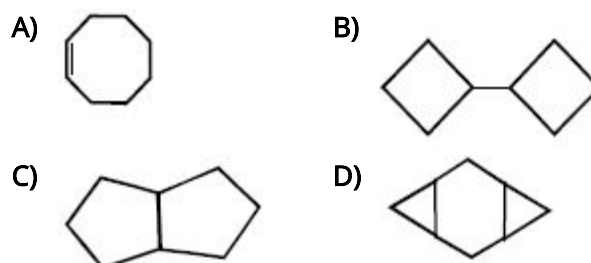
- A) Enantiomer B) Diastereomers
 C) Identical D) Structural isomer

78. The following compounds are identical with respect to



- A) molecular composition
 B) boiling point
 C) melting point
 D) IUPAC name

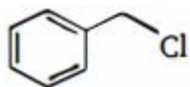
79. Which one of the compound is not isomer of others?



80. Nitroethane can exhibit one of the following kind of isomerism

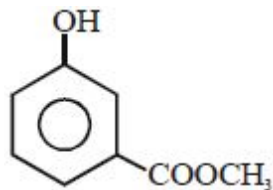
- A) Metamerism B) Optical activity
 C) Tautomerism D) Position isomerism

81. The *IUPAC* name of (figure) is



- A) Chloromethylbenzene
- B) Chlorophenylmethane
- C) (a) and (b) both
- D) none of these

82. The *IUPAC* name of (Image) is

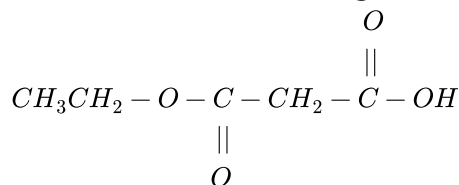


- A) 2-hydroxymethyl benzoate
- B) 2-hydroxyethyl benzoate
- C) Methyl 3-hydroxy benzoate
- D) 1-hydroxymethyl benzoate

83. The incorrect statement regarding ethyne is

- A) The C – C bonds in ethyne is shorter than that in ethene
- B) Both carbons are sp hybridised
- C) Ethyne is linear
- D) The carbon-carbon bonds in ethyne is weaker than that in ethene

84. The correct *IUPAC* name of given compound

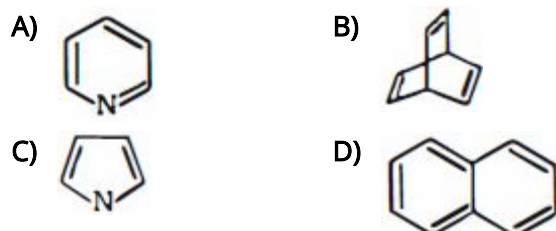


- A) Ethyl carboxylate ethanoic acid
- B) Ethoxy carbonyl ethanoic acid
- C) Ethyl –2- oxo propanoic acid
- D) 3- oxo –3- ethoxy propanoic acid

85. All the following *IUPAC* names are correct except :

- A) 1-chloro –1- ethoxy propane
- B) 1-amino –1- ethoxypropane
- C) 1-ethoxy –2- propanol
- D) 1-ethoxy –1- propanamine

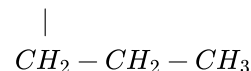
86. Which of the following compounds would not be considered aromatic in its behaviour?



87. In which of the following species is the underlined carbon having sp^3 hybridisation

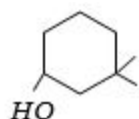
- A) $\text{CH}_3\text{C}\underline{\text{O}}\text{OH}$
- B) $\text{CH}_3\text{C}\underline{\text{H}}_2\text{OH}$
- C) $\text{CH}_3\text{C}\underline{\text{O}}\text{CH}_3$
- D) $\text{CH}_2 = \text{C}\underline{\text{H}} - \text{CH}_3$

88. *IUPAC* name of the following compound will be



- A) 3-ethyl –2- hexene
- B) 3-propyl –2- hexene
- C) 3-propyl –3- hexene
- D) 4-ethyl –4- hexene

89. The *IUPAC* name of the compound is



- A) 1,1-dimethyl–3-hydroxy cyclohexane
- B) 3,3- dimethyl–1-cyclohexanol
- C) 3,3-dimethyl–1-hydroxy cyclohexane
- D) 1,1-dimethyl–3-cyclohexanol

90. *IUPAC* name of the compound is $\text{CH}_3 - \text{CH}(\text{CH}_2\text{CH}_2\text{CH}_3) - \text{CH}_2 - \text{CH}(\text{OH}) - \text{CH}_3$ is

- A) 4-ethyl –2- pentanol
- B) 4-methyl –2- hexanol
- C) 2-ethyl –2- pentanol
- D) 3-methyl –2- hexanol

Biology - (Zoology)

91. Silencing of specific mRNA is possible via RNAi because of...

- A) Complementary dsRNA
- B) Inhibitory ssRNA
- C) Complementary tRNA
- D) Non-complementary ssRNA

92. Which of the following genetically engineered organisms was used by Eli Lilly to prepare human insulin?

- A) Bacterium
- B) Yeast
- C) Virus
- D) Phage

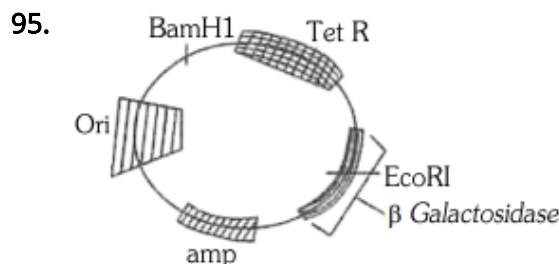
93. Polymerase chain reaction (PCR) amplifies DNA following the equation.

- A) N^2
- B) 2^n
- C) $2n + 1$
- D) $2 N^2$

94. Which of the following enzyme(s) are NOT essential for gene cloning?
- Restriction enzymes
 - DNA ligase
 - DNA mutase
 - DNA recombinase
 - DNA polymerase

Choose the correct answer from the options given below:

- | | |
|-----------------|-----------------|
| A) C and D only | B) A and B only |
| C) D and E only | D) B and C only |



In the above represented plasmid, an alien piece of DNA is inserted at EcoRI site. Which of the following strategies will be chosen to select the recombinant colonies?

- Using ampicillin & tetracycline containing medium plate.
- Blue color colonies will be selected.
- White color colonies will be selected.
- Blue color colonies grown on ampicillin plates can be selected.

96. The blue and white selectable markers have been developed which differentiate recombinant colonies from non-recombinant colonies on the basis of their ability to produce colour in the presence of a chromogenic substrate.

Given below are two statements about this method:

Statement I : The blue coloured colonies have DNA insert in the plasmid and they are identified as recombinant colonies.

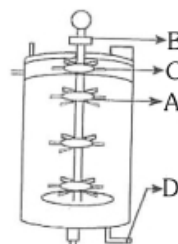
Statement II : The colonies without blue colour have DNA insert in the plasmid and are identified as recombinant colonies.

In the light of the above statements, choose the most appropriate answer from the options given below:

- Both Statement I and Statement II are correct
- Both Statement I and Statement II are incorrect

- Statement I is correct but Statement II is incorrect
- Statement I is incorrect but Statement II is correct

97. Identify the part of a bio-reactor which is used as a foam braker from the given figure.



- | | | | |
|------|------|------|------|
| A) A | B) B | C) D | D) C |
|------|------|------|------|

98. Given below are two statements :
Statement I: The DNA fragments extracted from gel electrophoresis can be used in construction of recombinant DNA.

Statement II : Smaller size DNA fragments are observed near anode while larger fragments are found near the wells in an agarose gel.

In the light of the above statements, choose the most appropriate answer from the options given below :

- Both statement I and statement II are correct
- Both statement I and statement II are incorrect
- Statement I is correct but statement II is incorrect
- Statement I is incorrect but statement II is correct

99. Which of the following is an example of non-distilled alcoholic beverage produced by yeast?

- | | | | |
|-----------|-----------|---------|--------|
| A) Whisky | B) Brandy | C) Beer | D) Rum |
|-----------|-----------|---------|--------|

100. Identify the statement that is NOT correct.

- Each antibody has two light and two heavy chains.
- The heavy and light chains are held together by disulfide bonds.
- Antigen binding site is located at C-terminal region of antibody molecules.
- Constant region of heavy and light chains are located at C-terminus of antibody molecules

101. Choose the true statements

I. Innate immunity is accomplished by providing different types of barriers

II. Acquired immunity is present from the birth and is inherited from parents

III. Sweat, tears, acid in the stomach and saliva prevent microbial growth

- A) I–True II–False III–True
- B) I–True II–True III–False
- C) I–False II–False III–True
- D) I–False II–True III–False

102. Modern lifestyle is responsible for

- A) Lowering of immunity
- B) More sensitivity of allergens
- C) Both (a) and (b)
- D) None of the above

103. Heroin is obtained from plant of family

- A) Papaveraceae B) Leguminosae
- C) Cruciferae D) Liliaceae

104. Which one of the following statements is correct?

- A) Patients, who had undergone surgery are given cannabinoids to relieve pain
- B) Benign tumors show the property of metastasis
- C) Heroin accelerates body functions
- D) Malignant tumors may exhibit metastasis

105. Nicotine intake stimulates the ...A... to release ...B... and ...C... into blood circulation. This leads to an increase in ...D... and an increase ...E... Here A and E refers to

- A) A–adrenal gland, B–adrenaline, C–nor-adrenaline, D–blood pressure, E– heart rate
- B) A–thyroid gland, B–thyroxine, C–parathyroxine, D–blood pressure, E– heart rate
- C) A–adrenal gland, B–thyroxine, C–nor-adrenaline, D–blood pressure, E–heart rate
- D) A–gonads, B–adrenaline, C–nor-adrenaline, D–blood pressure, E– heart rate

106. An autoimmune disease is

- A) AIDS B) Haemophilia
- C) Allergy D) Myastheniagravis

107. Match the following diseases with the causative organism and select the correct option.

Column –I	Column –II
(a) Typhoid	(i) Wuchereria
(b) Pneumonia	(ii) Plasmodium
(c) Filariasis	(iii) Salmonella
(d) Malaria	(iv) Haemophilus

(a) (b) (c) (d)

- A) (iv) (i) (ii) (iii)
- B) (i) (iii) (ii) (iv)
- C) (iii) (iv) (i) (ii)
- D) (ii) (i) (iii) (iv)

108. Which of the following disease caused internal bleeding, muscular pain, fever, anaemia and blockage of the intestinal passage?

- A) Ascariasis B) Filariasis
- C) Amoebiasis D) Trypanosomiasis

109. Antigen is a substance which

- A) lowers body temperature
- B) destroys harmful bacteria
- C) triggers the immune system
- D) is used as an antidote to poison

110. The tests that are used in the diagnosis of AIDS are

- A) ELISA and immunoblot
- B) Northern blot and ELISA
- C) ELISA and southern blot
- D) Western blot and ELISA

111. A - The vaccines generate memory - B and T - cells. R - In vaccination activated pathogens are introduced into the body.

- A) A and R both are correct.
- B) A and R both are incorrect.
- C) A is correct, R is incorrect.
- D) A is incorrect, R is correct.

112. Assertion : Cocaine has a potent stimulating action on central nervous system, producing a sense of euphoria and increased energy.

Reason : Injecting the microbes intentionally during immunisation or infectious organisms gaining access into body during natural infection induces active immunity.

- A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- C) If the Assertion is correct but Reason is incorrect.
- D) If both the Assertion and Reason are incorrect.

113. Which is the particular type of drug that is obtained from the plant whose one flowering branch is shown here?



- A) Hallucinogen B) Depressant
C) Stimulant D) Pain killer
114. Which one of the following non-infectious disease is the major cause of death in humans ?
- A) Cancer B) *AIDS* C) Asthma D) Typhoid

115. Anamnestic response

- A) Encounter a pathogen first time
B) Encounter pathogen second time
C) A and B both
D) None of the above

116. Symptoms of amoebiasis

- A) Abdominal pain, cramps, stools with excess mucous and blood clots and constipation
B) Internal bleeding, muscular pain, fever, anemia
C) Inflammation of legs, mutation in genital organs
D) Fever, chills, Headache, cough

117. Which of the following is included in biopesticide?

- A) Viruses and bacteria only
B) Viruses, bacteria and fungi only
C) Viruses, bacteria, fungi and protozoa
D) Viruses, bacteria, fungi and protozoa only

118. __A__ enzyme is used to remove clots from the blood vessels of patients who have undergone __B__ leading to heart attack.

- A) A—Streptokinase; B—Myocardial infarction
B) A—Lipases; B—Arteriosclerosis
C) A—Proteases; B—Myocardial infarction
D) A—Pectinases; B—Atherosclerosis

119. Organic farming includes

- I. use of biofertilisers and biopesticides
II. crop rotation
III. locally developed pest resistant varieties
Choose the correct option

- A) I and II
B) I and III
C) II and III
D) I, II and III

120. Yeast have been used for the commercial production of

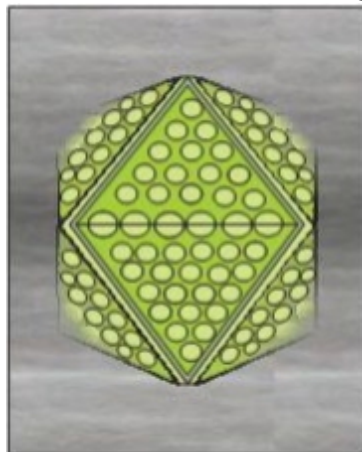
I. ethanol II. bread III. cheese

Choose the correct option

- A) I and II B) I and III
C) I, II and III D) None of these

121. Diagram of Adenovirus

Choose the correct for the given diagram



- A) Adenovirus - Cause of respiratory disease
B) Bacteriophage - Tobacco mosaic disease
C) Viroid - Potato spindle tumor disease
D) Prion - Alzheimer's disease

122. A patient brought to a hospital with myocardial infarction is normally immediately given

- A) penicillin B) streptokinase
C) cyclosporin- A D) statins.

123. Primary treatment is the

- A) physical removal of large and small particles from sewage
B) biological removal of large and small particles from sewage
C) Both (A) and (B)
D) chemical removal of large and small particles from sewage

124. Priyons are....

- A) Fungi with lipids B) Proteineous microbes
C) Algae with carbohydrates D) None

125. Insecticides kill

- A) Harmful insects
B) Only plant pests
C) Specific insects
D) Both harmful and useful insects

126. *VAM* represents

- A) Saprophytic fungi
B) Symbiotic fungi

- C) Saprophytic bacteria
- D) Symbiotic bacteria

127. 'ELISA' test is done for the diagnosis of

- A) Anthrax B) HIV C) Hepatitis D) Malaria

128. Chemicals, which can induce cancer are called

- A) Mutagenic agents and produce benign tumour
- B) Carcinogens and produce non-malignant tumour
- C) Mutagenic agents and do not produce malignant tumour
- D) Carcinogens and produce malignant tumour

129. Cell division or mitosis is a normal process in a living cell, but sudden and abnormal mitosis in an organ will frequently result in a

- A) Zygote B) Gastrula C) New organ D) Cancer organ

130. Man in the life cycle of Plasmodium is

- A) Primary host B) Secondary host
- C) Intermediate host D) None of these

131. Sensitivity to any allergen is related to

- A) Deviation from the process of immunity
- B) Age of the person
- C) Eating habit
- D) Rise in environmental temperature

132. Infectious or communicable diseases

- A) Cannot be communicated from one man to another
- B) Can develop infection from one man to another
- C) Can spread without any intermediate host
- D) Can spread with some intermediate host

133. Which one of the following sets includes the bacterial disease

- A) Cholera, typhoid, mumps
- B) Tetanus, tuberculosis, measles
- C) Malaria, mumps, poliomyelitis
- D) Diphtheria, leprosy, plague

134. The biological agents of disease include

- A) Minerals, vitamins, proteins and carbohydrates
- B) Viruses, bacteria, fungi, helminths and other organisms
- C) Heat, cold, humidity pressure, radiations
- D) All the above

135. Malarial parasites could be best obtained from a patient

- A) An hour before rise of temperature
- B) When temperature rises with rigor
- C) When temperature comes to normal
- D) Few hours after the temperature reaches to normal

Biology - (Botany)

136. Basmati is unique for its aroma and flavour, whose *A*... varieties are cultivated in *B*....

Here *A* and *B* refers to

- A) *A* – 27; *B*—America
- B) *A* – 30; *B*—America
- C) *A* – 27; *B*—India
- D) *A* – 30; *B*—India

137. With regard to insulin choose correct options.

- (a) *C*—peptide is not present in mature insulin.
 - (b) The insulin produced by *rDNA* technology has *C*—peptide.
 - (c) The pro-insulin has *C*—peptide.
 - (d) *A* —peptide and *B* —peptide of insulin are interconnected by disulphide bridges.
- Choose the correct answer from the options given below.

- A) (b) and (d) only
- B) (b) and (c) only
- C) (a), (c) and (d) only
- D) (a) and (d) only

138. Identify the incorrect statement.

- (i) *Bacillus thuringiensis* are available in sachets as dried spores which are mixed with kerosine and sprayed on to vulnerable plants
- (ii) *B. thuringiensis* are used to control butterfly caterpillars, but leave other insects unharmed
- (iii) The toxin of *B. thuringiensis* is released in the blood of larvae and the larvae get killed
- (iv) By the development of methods of genetic engineering *B. thuringiensis* toxin genes have been introduced

- A) (i) and (ii)
- B) (ii) and (iii)
- C) (i) and (iii)
- D) (ii) and (iv)

139. Which of the following is correctly matched?

- A) *Agrobacterium tumefaciens* - Tumour
- B) *Thermos aquaticus* - Bt-gene
- C) *pBR322* — Enzyme
- D) Ligase - Molecular scissors

140. Which one of the following molecular diagnostic technique is used to detect the presence of a pathogen in its early stage of infection?

- A) Angiography
- B) Radiography
- C) Enzyme replacement technique
- D) Polymerase Chain Reaction (*PCR*)

141. Silencing of a gene could be achieved by the use of

- A) *RNAi*
- B) Antisense *RNA*
- C) Both (a) and (b)
- D) None of these

142. Which of the following shows correct chronological order of the events occurring during callus culture?

- A) Callus → Cell division → Explant → Addition of cytokinin → Cells acquire meristematic property
- B) Explant → Callus → Cell division → Addition of cytokinin → Cells acquire meristematic property
- C) Explants → Cell division → Callus → Addition of cytokinin → Cells acquire meristematic property
- D) Callus → Explant → Cell division → Addition of cytokinin → Cells acquire meristematic property

143. A : Transgenic mice are being used to test the safety of the polio vaccine.

R : It could replace the use of monkeys to test the safety of batches of the vaccine.

- A) Assertion and Reason both are correct and also correct explanation.
- B) Assertion and Reason both are correct but not explanation of assertion.
- C) Assertion is correct, but Reason is incorrect.
- D) Both Assertion and Reason are incorrect.

144. A : *ELISA* can enable very easy detection of infection through antigen antibody interaction.

R : It is a nucleic acid based diagnostic tool that can confirm presence of infectious microbe at early stages.

- A) Assertion and Reason both are correct and also correct explanation.
- B) Assertion and Reason both are correct but not explanation of assertion.
- C) Assertion is correct, but Reason is incorrect.
- D) Both Assertion and Reason are incorrect.

145. For effective treatment of the disease, early diagnosis and understanding its pathophysiology is very important. Which of the following molecular diagnostic techniques is very useful for early detection?

- A) Western Blotting Technique

- B) Southern Blotting Technique
- C) *ELISA* Technique
- D) Hybridization Technique

146. Which of the following are fused in somatic hybridization involving two varieties of plants?

- A) Somatic embryos
- B) Protoplasts
- C) Pollens
- D) Callus

147. Consider the following statements about insulin.

I. Human insulin is made-up of 51 amino acids arranged in two polypeptide chains.

II. The two polypeptide chains are interconnected by two disulphide bridges.

III. In mammals including humans, insulin is synthesised as a pro-hormone, which contains an extra stretch called the C-peptide.

IV. C-peptide is not present in the mature insulin

Which of the statements given above are correct?

- A) I, II and III only
- B) I, II and IV only
- C) I, III and IV only
- D) I, II, III and IV

148. What triggers activation of protoxin to active *Bt* toxin of *Bacillus thuringiensis* in boll worm?

- A) Body temperature
- B) Moist surface of midgut
- C) Alkaline *pH* of gut
- D) Acidic *pH* of stomach

149. Silencing of *mRNA* has been used in producing transgenic plants resistant to

- A) bollworms
- B) nematodes
- C) white rusts
- D) bacterial blights.

150. Assertion : Insect resistant transgenic cotton has been produced by inserting *Bt* gene.

Reason : The *Bt* gene is derived from a bacterium.

- A) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.
- B) If both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.
- C) If the Assertion is correct but Reason is incorrect.
- D) If both the Assertion and Reason are incorrect.

151. GMO technology is useful for

- A) Making crops more tolerant to abiotic stresses

- B) Helping to reduce post-harvest losses
- C) Enhancing nutritional value of food
- D) All the above

152. Which body of the Government of India regulates *GM* research and safety of introducing *GM* organisms for public services ?

- A) Genetic Engineering Approval Committee
- B) Research Committee on Genetic Manipulation
- C) Bio-safety committee
- D) Indian Council of Agricultural Research

153. Genetic engineering is the

- A) Formation of new gene artificially
- B) Formation of *RNA* from *DNA* artificially
- C) Modification of genes artificially
- D) Formation of *DNA* from non *DNA* material

154. A..... infects the roots of tobacco plants which reduce the production of tobacco.

- A) Nematode (*Meloidogyne incognita*)
- B) Coleopterans (beetles)
- C) Lepidopterns (armyworm)
- D) Dipterans (Mosquitoes)

155. The crops engineered for glyphosate are resistant/tolerant to

- A) insects B) herbicides C) fungi D) bacteria.

156. It is now possible to breed plants and animals with desired characters through

- A) Genetic engineering
- B) Chromosome engineering
- C) Ikebana technique
- D) Tissue culture

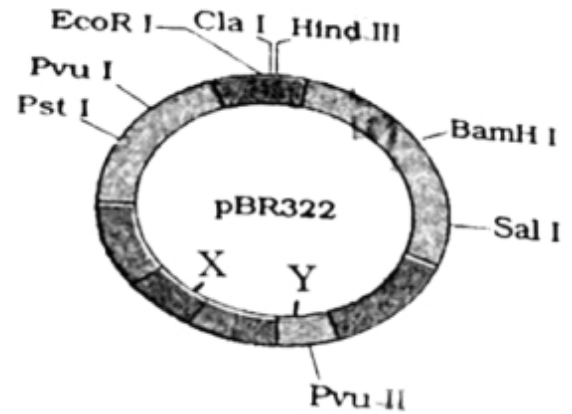
157. *Bacillus thuringiensis* is a good

- A) Biofertilizer B) Biopesticide
- C) Biofuel D) Single cell protein

158. The "Ti plasmid" of *Agrobacterium tumefaciens* stands for

- A) Tumor independent plasmid
- B) Tumor inducing plasmid
- C) Temperature independent plasmid
- D) Tumour inhibiting plasmid

159. The following diagram showing restriction sites in *E. coli* cloning vector pBR322. Find the role of 'X' and 'Y' genes:



- A) The gene 'X' is responsible for controlling the copy number of the linked DNA and 'Y' for protein involved in the replication of Plasmid.
- B) The gene 'X' is for protein involved in replication of Plasmid and 'Y' for resistance to antibiotics.
- C) Gene 'X' is responsible for recognitions sites and 'Y' is responsible for antibiotic resistance
- D) The gene 'X' is responsible for resistance to antibiotics and 'Y' for protein involved in the replication of Plasmid.

160. Natural genetic engineer is:

- A) *Bacillus subtilis*
- B) *Pseudomonas* spp
- C) *Escherichia coli*
- D) *Agrobacterium tumefaciens*

161. Cutting of a piece of *DNA* from a plasmid was done with the help of ...A... enzymes, popularly known as ...B.... Here A and B can be

- A) A—*Tu* ligases; B— Molecular glu
- B) A—Restriction enzyme; B— Molecular scissors
- C) A—Joining enzyme; B— Molecular glu
- D) A — *DNA* polymerases; B—Synthesising enzymes

162. In bacteria, genes for antibiotic resistance are usually located in:

- A) Chromosomal DNA
- B) Cytoplasm
- C) Mitochondria
- D) Plasmids

163. Which cannot be used for direct gene transfer

- A) Biolistics (gene gun)
- B) Microinjection
- C) Electroporation
- D) *Agrobacterium tumefaciens*

164. Choose the correct pair from the following

- A) Exonucleases : Make cuts at specific positions within *DNA*
 B) Ligases : Join the two *DNA* molecules
 C) Polymerases : Break the *DNA* into fragments
 D) Nucleases : Separate the two strands of *DNA*
165. Due to chloramphenicol resistance gene, one is able to select a transformed cell in the presence of chloramphenicol. The chloramphenicol resistance gene in this case is called
- A) Origin of replication
 B) Selectable marker
 C) Cloning sites
 D) Insertional inactivation
166. *DNA* can be introduced into any cell by:
- A) Injection
 B) Being complexed with calcium salts
 C) Being placed along with the cell into a gene gun
 D) Gel electrophoresis
167. In recombinant *DNA* technique, the term vector refers to a
- A) Donor *DNA*, it is identified and picked up through electrophoresis
 B) Plasmid, transfers *DNA* into living cell
 C) Collection of entire genome in form of plasmid
 D) Enzyme, cuts the *DNA* at specific sites
168. Which of the following steps is/are catalysed by Taq polymerase in a *PCR*?
- A) Denaturation of template *DNA*
 B) Annealing of primers to template *DNA*
 C) Extension of primer end on template *DNA*
 D) All of these
169. *RNA* is removed by the treatment with
- A) Ribonuclease B) Protease
 C) Chitinase D) Cellulase
170. The components of a bioreactor are
- I. an agitator system
 II. an oxygen delivery system
 III. foam control system
 IV. temperature control system
 V. *pH* control system
 VI. sampling ports to with draw cultures periodically
 Choose the correct option
- A) I, II, III, IV and V
 B) II, IV, V and VI
 C) I, II, III, IV and VI
 D) All of these
171. Gene for cloning may be chemically synthesized:
- A) When the exact sequence of nucleotides is known
 B) Through the use of restriction enzymes and gel electrophoresis to separate restriction fragments
 C) By the Sanger method
 D) By making complementary *DNA* from genes without introns
172. Which of the following statement is not correct regarding *Eco RI* restriction endonuclease enzyme?
- I. *Eco. RI* restriction endonuclease enzyme is isolated from *Escherichia coli RY13*
 II. Its recognition sequence is $5' - GAATTC - 3'$
 $3' - CTTAAG - 5'$
 $5' - G - A - A - T - T - C - 3'$
 $3' - C - T - T - A - A - G - 5'$
 III. Its site of cleavage is
- A) I and II
 B) I and III
 C) I, II and III
 D) None of the above
173. Which of the following is the most accepted definition of biotechnology by European Federation of Biotechnology?
- A) Maintenance of sterile ambience for enabling growth of desired microbe/eukaryotic cell in large quantities
 B) Technique of using live organism or enzyme from organisms to produce products and processes useful to animals
 C) Process which use genetically engineered animals only on a large scale for benefit of mankind
 D) The integration of natural science and organisms cells, parts thereof and molecular analogues for products and services
174. Which one of the following statement is not true regarding gel electrophoresis technique?
- A) The separated *DNA* fragments are stained by using ethidium bromide.
 B) The presence of chromogenic substrate gives blue coloured *DNA* bands on the gel.
 C) Bright orange coloured bands of *DNA* can be observed in the gel when exposed to *UV* light.
 D) The process of extraction of separated *DNA* strands from gel is called elution.
175. Most suitable method of introducing alien *DNA* into a plant cell is

- A) Biolistics
- B) Micro-injection
- C) Lipofection
- D) Heat shock method

176. The specific palindromic sequence which is recognized by *EcoRI* is :

- A) 5' – GGATCC – 3'
3' – CCTAGG – 5'
- B) 5' – GAATTC – 3'
3' – CTTAAG – 5'
- C) 5' – GGAACC – 3'
3' – CCTTGG – 5'
- D) 5' – CTTAAG – 3'
3' – GAATTC – 5'

177. I. *DNA* being a hydrophilic molecule cannot pass through cell membranes.
II. The bacteria should be made competent to accept the *DNA* molecule.
The Correct option regarding the above statements is

- A) I is true, but II is false
- B) II is true, but I is false
- C) I and II are true
- D) I and II are false

178. *Taq* polymerase is isolated from a

- A) Virus
- B) Bacteria
- C) Fungus
- D) Protozoans

179. Recombinant *DNA* technology was discovered by

- A) Khorana
- B) Bateson and de Vries
- C) Watson
- D) Cohen and Boyer

180. Restriction endonucleases are

- A) Synthesized by bacteria
- B) Present in mammalian cells for degradation of *DNA*
- C) Used in genetic engineering
- D) Used for in vitro *DNA* synthesis